Contents

Exponential RV (cont'd) & Gamma RV

EEE 554 — Lecture #4

Exponential RV (continued)

Note that for an exponential random variable with parameter λ, the mean is $\mu_x = 1/\lambda$. The probability that the variable exceeds its mean is:

$$P(x > \mu_x) = \int_{1/\lambda}^{\infty} \lambda e^{-\lambda x} dx = e^{-1} \approx 0.37$$

Example (Markov Inequality): With x being exponential with $\lambda = 1$ (so $E[x] = 1$):

$$P\left(x \geq \frac{3}{2}\right) = \int_{3/2}^{\infty} e^{-x} dx = e^{-3/2} \approx 0.22$$

Using Markov's inequality to bound this probability:

$$P\left(x \geq \frac{3}{2}\right) \leq \frac{E[x]}{(3/2)} = \frac{1}{3/2} = \frac{2}{3} \approx 0.67$$

Gamma RV: $X \sim \Gamma(\alpha, \beta)$

Parameters: $\alpha > 0$ ("shape"), $\beta > 0$ ("rate")

PDF:

$$f_x(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

In this expression, Γ denotes the **gamma function**, defined for any complex number t that is not a non-positive integer by:

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} dx$$

Note that:

$$\Gamma(1) = \int_0^{\infty} x^0 e^{-x} dx = 1$$

And for $k = 1, 2, 3, \ldots$, using integration by parts:

$$\begin{aligned} \Gamma(k+1) &= \int_0^{\infty} x^k e^{-x} dx \\ &= -x^k e^{-x} \Big|_0^{\infty} + \int_0^{\infty} k x^{k-1} e^{-x} dx \\ &= k\Gamma(k) \end{aligned}$$

So, $\Gamma(2) = 1 \cdot \Gamma(1) = 1$, and $\Gamma(3) = 2 \cdot \Gamma(2) = 2$. In general, for any positive integer n:

$$\Gamma(n) = (n-1)!$$

Proof of Unit Integral for Gamma PDF:

$$\int_{-\infty}^{\infty} f_x(x)dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \beta^\alpha x^{\alpha-1} e^{-\beta x} dx$$

Let $u = \beta x$, so $du = \beta dx$. Substituting these in:

$$= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} u^{\alpha-1} e^{-u} du = \frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1$$

Moments: Similar calculations yield the mean and variance:

$$E[X] = \frac{\alpha}{\beta} \quad \text{and} \quad \text{var}(X) = \frac{\alpha}{\beta^2}$$

Gaussian CDF

Let $X \sim N(0,1)$. Then its Cumulative Distribution Function (CDF) is:

$$F_x(u) = \int_{-\infty}^{u} f_x(x)dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{u} e^{-x^2/2} dx = \Phi(u)$$

No closed-form expression for $\Phi(u)$ exists, but it can be evaluated numerically and is tabulated in many references. One often sees the complementary CDF (Q-function) defined as:

$$Q(u) \triangleq 1 - \Phi(u)$$

If $X \sim N(\mu, \sigma^2)$, then:

$$P(X \leq u) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{u} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Using the substitution $v = \frac{x-\mu}{\sigma}$, which implies $dv = \frac{dx}{\sigma}$:

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{u-\mu}{\sigma}} e^{-\frac{v^2}{2}} dv = \Phi\left(\frac{u-\mu}{\sigma}\right)$$

Example: Let $X \sim N(2,9)$. Find $P(|X-2| \geq 1)$. Because the distribution is symmetric around the mean ($\mu = 2$) and $\sigma = 3$:

$$\begin{aligned} P(|X-2| \geq 1) &= P(X \geq 3) + P(X \leq 1) \\ &= 2P(X \leq 1) \\ &= 2\Phi\left(\frac{1-2}{3}\right) = 2\Phi\left(-\frac{1}{3}\right) \end{aligned}$$